THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ray of light falling on a surface is the: 1

(a) normal **(b)** incident ray **(c)** emergent ray **(d)** reflected ray

2. The distance of the image from a plane mirror is ___ the distance of the object. 1

(a) unrelated to **(b)** double **(c)** half of **(d)** equal to

3. Which of these is non-luminous? 1

(a) the Moon **(b)** a bulb **(c)** a candle flame **(d)** a star

4. The partially dark region around the umbra is the: 1

(a) penumbra **(b)** image **(c)** umbra **(d)** focus

5. The ray that bounces back is the: 1

(a) incident ray **(b)** normal **(c)** refracted ray **(d)** reflected ray

6. In a plane mirror, the left and right appear: 1

(a) upside down **(b)** interchanged (laterally inverted)
(c) rotated **(d)** unchanged

7. Materials through which light passes completely are: 1

(a) translucent **(b)** opaque **(c)** luminous **(d)** transparent

8. A pinhole camera forms an image that is: 1

(a) the same **(b)** virtual **(c)** inverted **(d)** erect

9. The line at right angles to the reflecting surface is the: 1

(a) axis **(b)** normal **(c)** incident ray **(d)** tangent

10. The word AMBULANCE is written laterally inverted on vehicles so that: 1

(a) it hides
(b) it reads correctly in the rear-view mirror
(c) it is easy to paint
(d) it looks nice

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is lateral inversion? 2

12. What is the angle of reflection if the angle of incidence is 45°? 2

13. How is a shadow formed? 2

14. Write two properties of the image in a plane mirror. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the construction of a pinhole camera. 3

16. Why are the rear-view mirrors and the front of ambulances used with reversed text? 3

17. Why does the shadow of a person change in length during the day? 3

OR

Write the properties of the image formed by a pinhole camera. 3

18. Differentiate between regular and diffused reflection. 3

OR

Explain the reflection from water surface. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State the laws of reflection and describe an experiment to verify them. 5

OR

Explain how eclipses and the phases of Moon are related to shadows (qualitatively). 5

20. Explain the properties of the image formed by a plane mirror with the example of a person standing 3 m from it. 5

OR

Describe the images in a mirror and the use of mirrors in everyday life. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND